

Adv C Module

1. Subjective

2.1 Basic Refreshers

1. Where does the execution of a C program start? Explain the flow before main().
2. What are pre-increment and post-increment? Explain with examples.
3. What is conditional compilation? Give examples.
4. What are data types?

2.2 1D Pointers and Functions

1. What is the difference between pass by value and pass by reference?
2. What is a pointer?

2.3 Storage classes and memory segments

1. Explain different storage classes.
2. Explain the memory segments (text, data, bss, heap, stack).
3. What is a static variable? Difference between static local and static global?
4. Can a static variable be accessed in another file? Explain.
5. What happens if an extern variable is used without a previous declaration?
6. What error occurs in that case?
7. What happens if a static variable is declared inside a function?
8. How many times is it initialised if the function is called multiple times?

2.4 2D Pointers and DMA

1. What are the different types of pointers? Explain one by one.
2. What is a memory leak? How can it be avoided?
3. Explain malloc(), calloc(), and realloc().
4. Explain the function pointers.
5. What is DMA?
6. What is the difference between `const int *p` and `int const *p`?
7. Declare a constant variable and a constant pointer.

2.5 Preprocessing

1. What is optimisation?
2. What are the stages of compilation?
3. Explain the compilation process.

2.6 UDT

1. What is a structure?
2. Write a structure with three members.
3. What is the difference between a structure and a union?
4. How is data shared between members in a union?
5. How do you find the size of a structure?
6. What is structure padding?
7. How can you avoid structure padding when memory is limited?
8. What is #pragma pack? How is it related to padding?
9. Explain endianness using a union.
10. What is a bitfield? Where is it used?

2.7 Miscellaneous

1. Explain the 'volatile' keyword.

2.8 FILE I/O

1. A file contains numbers in this format

```
1 2 3 4
5 6 7 8
```

Without knowing how many numbers are in a line, read the file dynamically and store the numbers in a 2D array. Numbers in each line must be stored in a separate row.

2. Write a program for file handling to read, filter, and validate email IDs from a file using regex.

2. Programming

1. Write a program to reverse a string using pointers.
2. Write a program to find the second largest element in an array.
3. Write a program to reverse a number.
4. Write a program to check whether a number is prime.
5. WAP to find if a number is a power of 2 (with and without bitwise).
6. WAP to swap the kth and nth bits in a number and toggle them.

7. Write a bitwise program to generate an output number by right-shifting based on a position.
8. Write a program to add two numbers using a function pointer.
9. Write a macro to swap two variables.
10. Write a program to check whether a string is a palindrome.
11. WAP to convert input 0x12345 to output 0x54231.
12. WAP to swap the nibbles of a byte.
13. WAP to add two numbers using pointers.

MC Module

1. Basics

1. What is the difference between a microcontroller and a microprocessor?
2. What is memory-mapped I/O?
3. What is a bootloader? Explain the booting process.
4. What are GPIO pins?
5. What are volatile and non-volatile memories? Why is the volatile keyword required?
6. Explain Von Neumann and Harvard architectures.
7. How can you reduce power consumption in embedded systems?
8. Explain pull-up and pull-down resistors and their applications.
9. Explain cache types and memory hierarchy.
10. What is the difference between a 16-bit system and a 12-bit system?
11. How do you generate a 1 ms delay using a timer?
12. What is the size of an int in a PIC microcontroller?

2. Interrupts

1. What are interrupts?
2. What is interrupt latency? How can it be reduced?
3. Explain interrupt service routine (ISR) and timers.

3. Projects

1. Explain the car black box project.
2. Explain the CAN-based automotive dashboard project.

4. ADC

1. Explain ADC.
2. What is the difference between analogue and digital signals?
3. Why do we need an ADC in embedded systems?

5. Embedded Systems

1. What is an embedded system? Give examples.

6. Protocols

1. What is UART and how is it used in embedded systems?
2. How does UART communication work?
3. How do you send 10 bytes of data over UART?
4. Explain CAN protocol. Describe arbitration in CAN frames.
5. Explain the SPI protocol with a frame format.
6. Explain the I2C protocol with a frame format.

7. PWM

1. What is PWM frequency?
2. What is the PWM period?
3. What are ON time and OFF time in PWM?
4. What is the difference between PWM and DAC?
5. Where is PWM commonly used?
6. What happens if the duty cycle is 0% or 100%?
7. What is the average voltage of a PWM signal?

CPP Module

1. What is the difference between function overloading and operator overloading?
2. Explain the concepts of OOP.
3. What is an abstract class?
4. Explain the differences between class and structure.
5. Explain function overloading.

Ds Module

1. Basics

1. What is the difference between an array and a linked list?
2. Explain time complexity.

2. Linked lists

1. Write a program to print elements of a singly linked list.
2. Write a program to detect a loop in a linked list.
3. Write a program to traverse a circular linked list.
4. WAP to print a reversed linked list without actually reversing it.
5. Write a program to create a linked list, insert elements, find the middle node, and insert a new node at the middle position.

3. Stack

1. WAP to check whether the stack grows upward or downward.
2. Explain LIFO with a real-time example.
3. What are the basic stack operations?
4. What is stack overflow and stack underflow?

4. Searching and Sorting Techniques

1. What is the difference between Bubble Sort and Selection Sort?
2. Time complexity of linear search (best, average, worst).
3. What is the difference between iterative and recursive binary search?
4. Why is binary search faster than linear search?

5. Queue

1. Explain the working of a stack and queue with real-life examples.
2. Explain FIFO with a real-time example.
3. What is the producer-consumer problem?
4. How is a queue used in scheduling?

6. Hashing

1. What is hashing? And hash table.
2. What is a hash function?

7. Trees

1. Explain binary search tree. Write code for BST operations.
2. WAP to print the leftmost node at each level of a binary tree.

LI Module

1. Basics

1. Explain CPU scheduling mechanisms.
2. What is an operating system?
3. What is the difference between OS and RTOS? Give examples.
4. What resources does an OS manage?
5. What is a kernel? What does a kernel do?
6. Explain the Linux boot-up process.
7. What is the difference between user space and kernel space?
8. How does communication happen between user space and kernel space?

2. System call

1. What is a system call?
2. Why do we need system calls?
3. How does a system call work internally?
4. What is the difference between a function call and a system call?

3. Networking

1. What is a network protocol?
2. What are the layers of the OSI model?
3. What is an IP address?
4. What is the difference between IPv4 and IPv6?
5. What is a port number?
6. What is the difference between a MAC address and an IP address?

4. Process

1. How do you see running processes in Linux?
2. What is a PID?
3. How do you find the current process ID?
4. How do you terminate a process?
5. What command is used to kill a process?

5. IPC

1. What is IPC? Give examples.
2. What are the types of IPC?
3. What does a pipe do?
4. How do you declare and use a pipe?

6. Signal

1. What is a signal in Linux?
2. Why are signals used?
3. What are synchronous and asynchronous signals?
4. What is signal handling?
5. How do you catch a signal?
6. What is the difference between `signal()` and `sigaction()`?

7. Socket

1. What is a socket?
2. What are the types of sockets?
3. Which system call is used to create a socket?
4. What is the difference between `send()` and `write()`?
5. What is the difference between `recv()` and `read()`?
6. What is the difference between `select`, `poll`, and `epoll`?

8. Threads

1. What is the difference between a process and a thread with real-world examples?
2. Explain threads and processes in detail.
3. What is multithreading, and where is it used?
4. Explain mutexes and semaphores with real examples.
5. What are the possibilities of deadlocks?
6. What are synchronisation mechanisms?

9. Process and Memory Management

1. What are paging, page fault, and demand paging?
2. What is memory management?
3. Why is memory management important?
4. What are the types of memory?
5. What is the difference between volatile and non-volatile memory?
6. What is primary memory?
7. What is secondary memory?

General Questions

1. Have you done troubleshooting? Explain any troubleshooting experience.
2. You have a watch—how will you test it? (Choose analog or digital.)
3. If you alone have to test thousands of watches, how will you plan the testing?
4. Write test cases to verify a function that copies an array.
5. How would you test a system designed to train an AI model for a self-driving car?
6. What will you check when a sensor is not sending data to a microcontroller?
7. What will you check when a new microcontroller is given to you?
8. What is chmod and its use?
9. How do Python and C differ from a computer architecture perspective?
10. What are Python testing frameworks? Explain with examples.
11. Platform-based questions: ARM, x86, Windows, Linux, macOS.
12. How do you change file permissions?
13. How do you find a file?
14. Explain Modbus RTU.
15. How did you interface STM32 with Modbus RTU?
16. How did you connect STM32 with a Schneider Electric M580 PLC?